

Prasad Suryabhan Gite

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Machine Learning Engineer | Full Stack Developer

PROFESSIONAL EXPERIENCE

Lead Machine Learning Engineer

09/2025 – 02/2026 | Remote

Aurum Innovations

- Designed and deployed scalable computer vision pipelines for automated car damage detection, reducing manual inspection effort by 60%.
- Performed feature engineering and data preprocessing on 12,000+ annotated images across 5 classes to improve model generalization. Trained YOLOv8 models achieving 91% mAP@0.5; improved recall by 18% using class balancing and focal loss.
- Built data pipelines for image augmentation and preprocessing, reducing training time by 35% using CUDA-based parallel processing.
- Optimized model inference using TensorRT and ONNX Runtime, enabling low-latency real-time predictions in production.
- Deployed ML models in production environments with REST APIs and integrated with backend systems.
- Led end-to-end ML lifecycle including data collection, model training, evaluation, deployment, and monitoring.

AI/ML Research Intern

07/2024 – 12/2024

Research Internship

- Developed NLP-based chatbot using Retrieval-Augmented Generation (RAG) and LLMs, improving response accuracy by 25%.
- Built semantic search system using vector databases for efficient document retrieval, reducing latency by 40%.
- Implemented intent classification models achieving 88% accuracy across 12 categories.
- Designed data pipelines for document ingestion, embedding generation, and query processing.
- Collaborated in an agile team environment contributing to system design, experimentation, and evaluation.

SKILLS

Languages

Python, C++

AI/ML

PyTorch, YOLOv8, TensorRT, ONNX Runtime, OpenCV, Scikit-Learn, LSTM, ARIMA, RAG, LLMs, Fine-tuning, Pandas, NumPy

Databases

MongoDB, DBMS, PostgreSQL, SQL

MLOps & Cloud

AWS (EC2, S3), Docker, Git, Jupyter Notebook, CUDA, GPU-accelerated training

Backend & Systems

Node.js, Express.js, REST APIs, WebSockets, Service-Oriented Architecture

Others

LangChain, Vector Databases, CI/CD, DSA, OOP, Computer Networks, Dijkstra's Algorithm

PROJECTS

FlowVest: AI-Powered Financial Management Platform for Small Businesses(Ongoing)

React.js, WebSocket, Node.js, MongoDB, LSTM, ARIMA, Banking API(Plaid), Quickbooks, AWS

- Built a real-time cash flow monitoring platform for SMBs integrating Plaid and QuickBooks APIs; deployed on AWS (EC2, S3) with containerized Node.js microservices.
- Implemented LSTM and ARIMA time-series models for cash flow forecasting; backtested on 24-month synthetic datasets with RMSE 8% lower than baseline ARIMA-only model.
- Implemented responsive React.js UI with WebSocket-driven live updates, reducing page load times by 30%.

SmartPath: Dynamic Routing System for Warehouse Optimization

MongoDB, Express.js, React.js, Node.js, Dijkstra's Algorithm, Websockets

- Developed a real-time warehouse routing platform; implemented Dijkstra's Algorithm for dynamic pathfinding, reducing routing errors by 30% vs. static routing.
- Integrated Arduino hardware via REST API for real-world path adjustment, bridging software and IoT layers.
- Designed a WebSocket-powered live dashboard enabling warehouse operators to monitor 50+ routing events per minute.

AgroVision: Thermal Imaging Drone for Crop Health Monitoring

Thermal Imaging Camera, Python(OpenCV), MATLAB(Image Analysis), ML models(Predictive Crop Health)

- Built a UAV-mounted thermal imaging system using OpenCV and MATLAB to detect early-stage crop stress across multi-hectare farm plots.
- Trained predictive ML models on thermal image datasets to classify crop health with 84% accuracy across 4 stress categories.
- Handled end-to-end hardware-software integration including sensor calibration, image capture pipeline, and data visualization.

EDUCATION

Vishwakarma Institute of technology, Pune

2021 – 2025 | Pune

Bachelor of Technology

Army Public School Devlali

Nashik

SSC, HSC

CERTIFICATES

- Mastering DSA with C++
- IBM Data Science Professional Certificate - Coursera
- Full Stack Web Development Certification
- Generative AI for Everyone - Coursera
- Database Engineering(RDBMS)
- Certificate in English Comprehensive Communication

ACHIEVEMENTS & EXTRACURRICULAR

- Qualified for **Smart India Hackathon (SIH)** internal rounds in 2023 and 2024, leading SDE-focused solution teams.
- Participated in **Citi Bank Inter-College Hackathon**, developed a backend for fintech case study (shortlisted in internal round).
- Volunteered with the Social Welfare and Development Society to assist Maharashtra Police in traffic management and public safety initiatives.